

Computer Networks

Project Report

NASA Apache Log Summarizer



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Chapter 1: Introduction and Scope

1.1 Problem, Solution, and Relevance

Problem: Manually analysing millions of lines of unstructured web server logs (such as the NASA KSC data) is impossible.

Solution: We created an **Automated Python Tool** to process this data, focusing on **Network Automation** and **Intelligent Management**.

Objectives & Tools:

- **Data Extraction (Parsing):** Use Regex for the Common Log Format (CLF).
- **Data Analysis (Metrics):** Use Pandas for scalability, calculating hourly traffic, error rates, and Top IPs/URLs.
- **Visualisation & Report:** Use Matplotlib to generate charts and compile them into an HTML Report.

1.2 Ethical Statement

All project work is original and thoroughly tested. We acknowledge using AI assistance for initial code structuring, with all final code reviewed and verified by us, in compliance with all ethical requirements.

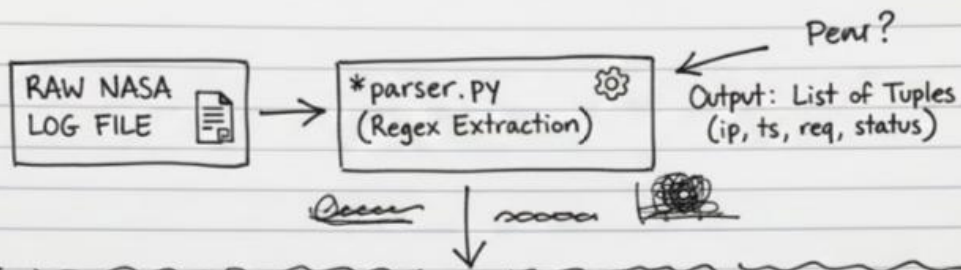
Chapter 2: Technical Design and Implementation

2.1 Modular System Architecture

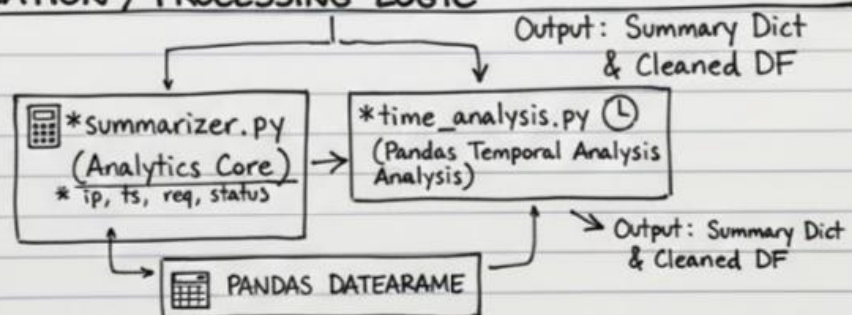
The project is built on a clear, four-stage pipeline for modularity and easy debugging

SYSTEM ARCHITECTURE - NASA LOG ANALYZER

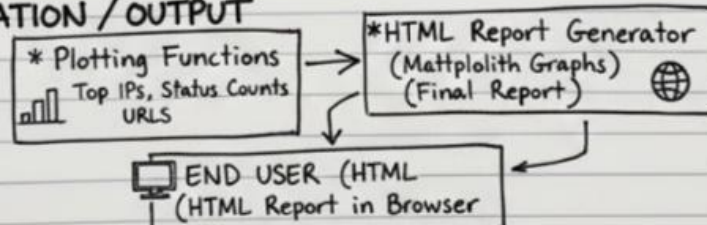
1. DATA INGESTION / PARSING



2. APPLICATION / PROCESSING LOGIC



3. PRESENTATION / OUTPUT



Stage	Module	Function
Ingestion & Parsing	parser.py	Finds log file and uses Regex to break raw text into structured records (IP, Time, Status).
Analysis	time_analysis.py	Uses Pandas for efficient calculations like Time-Series Resampling (resample('H')) and filtering Status Code >=400.
	error_analysis.py	
Reporting	report_generator.py	Generates Matplotlib charts and compiles all data into the final HTML file.

2.2 Core Technical Depth

- Regex Pattern:** We apply the CLF-specific pattern

`r'(\d+\.\d+\.\d+\.\d+)\s-\s-\s\[([.]*?)\]\s"([.]*?)"\s(\d{3})'`

to ensure technical rigor in data extraction.

- Scalable Analysis (Pandas):** Instead of slow manual loops, we use Pandas methods:
 - Temporal Analysis:** `df.resample('H').size()` for calculating requests per hour.
 - Error Analysis:** Filtering (`df[df["status"] >= 400]`) and counting (`value_counts()`) to isolate IPs causing errors for security auditing.
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Chapter 3: Results and Visualization

This chapter presents the quantitative and visual output of the working prototype.

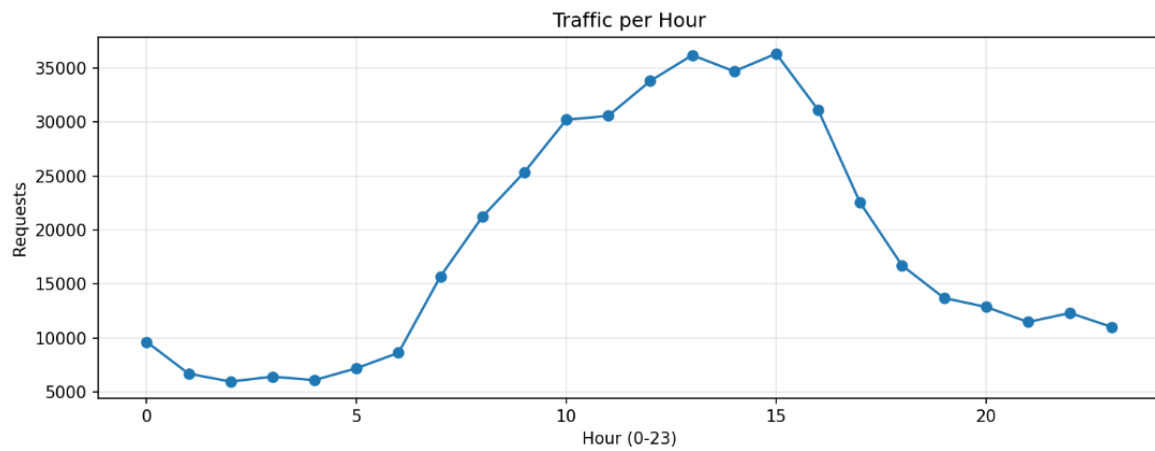
3.1 Key Metrics

Metric	Result	Insight
Total Requests	446,494	Total server load for the analysed period.
Unique Client IPs	23,210	Size of the unique audience.
Success Rate (200)	396,297	High volume of successful transactions.
404 Not Found Errors	2,272	Indicates broken links on the website (404 status).
Top Requested URL	/images/NASA-logosmall.gif	The single most popular resource on the server.

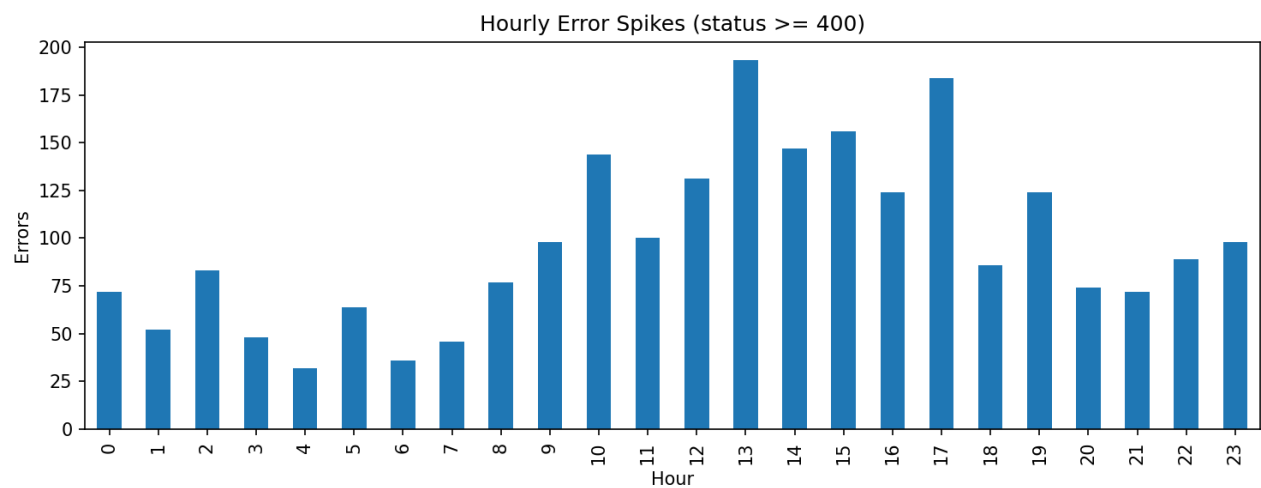
3.2 Graphical Interpretation

The generated charts provide crucial insights for Network Management:

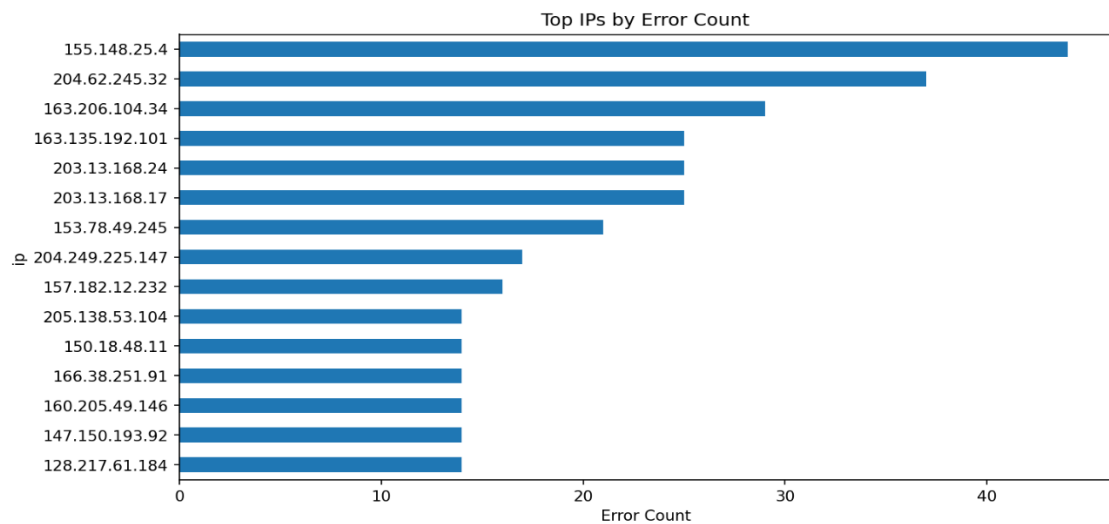
- **Traffic per Hour:** The line plot shows the busiest hours (Temporal Trend).



- **Hourly Error Spikes:** The bar plot isolates and tracks Status ≥ 400 errors by hour.



- **Top IPs by Error Count:** This chart instantly reveals the **most problematic client IPs** for security action (e.g., IP **155.148.25.4** generated the highest count of requests/errors).



More Visuals are added in the HTML Report file and in the folder as well.

Chapter 4: Conclusion and Future Scope

4.1 Conclusion

The project successfully delivered a robust and modular solution that converts complex network data into visual, actionable intelligence. We demonstrated expertise in **Regex**, **Pandas data manipulation**, and **Matplotlib visualisation**, fulfilling all technical and demonstrative requirements.

4.2 Future Scope

1. **Bandwidth Integration:** Integrate the size field (bytes sent) to calculate and visualize actual network bandwidth usage (MB/GB).
 2. **Geolocalization:** Add a module to map the Top Error IPs to geographical locations for enhanced security analysis.
 3. **Real-Time Capability:** Adapt the system for near real-time processing of incoming logs.
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